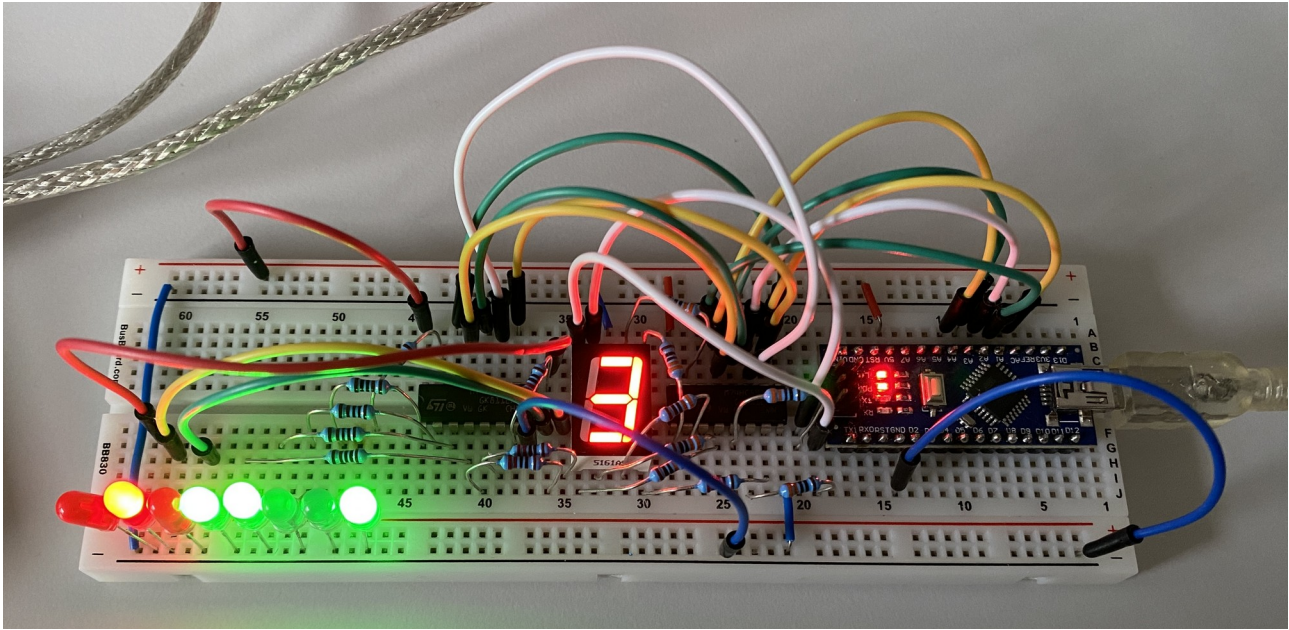




```

/* ET_49_74HC595_SPI_16bit
 *
 * Pinbelegung
 *
 * Funktion | M74HC595B1 | Arduino | MC
 * Neg CLR  | 10          | A1       | PC0
 * Data     | 14          | A2       | PC1
 * SRCLK    | 11          | A3       | PC2
 * RCLK     | 12          | A4       | PC4
 * Neg OE   | 13          | A5       | PC5
 *
 * LA steht für Low Aktiv.
 */

#define PIN_CLR_LA 0
#define PIN_DATA 1
#define PIN_CLK 2
#define PIN_RCLK 3
#define PIN_OE_LA 4

#define THIRD_CLK_MS 10

```

```

// Aufgabe 1: (0x37 = 0011 0111)
// const uint16_t data = 0x0037;

// QI und QII ansteuern (0x594F = 0101 1001 0100 1111)
const uint16_t data = 0x594F;

void transmit_data() {
    PORTC &= ~(1 << PIN_CLR_LA);
    delay(THIRD_CLK_MS);

    PORTC |= 1 << PIN_CLR_LA;
    delay(THIRD_CLK_MS);

    for (int8_t bit_index = 16; bit_index >= 0; bit_index--) {
        // Serial data bit
        const byte data_bit = (data >> bit_index) & 0x01;

        // Clear PIN_DATA then set the data_bit
        PORTC = (PORTC & ~(1 << PIN_DATA)) | data_bit << PIN_DATA;
        delay(THIRD_CLK_MS);

        // Clock high
        PORTC |= 1 << PIN_CLK;
        Serial.println("Clock high");
        delay(THIRD_CLK_MS);

        // Clock low
        PORTC &= ~(1 << PIN_CLK);
        Serial.println("Clock low");
        delay(THIRD_CLK_MS);
    }
}

```

```

    PORTC |= 1 << PIN_RCLK;
    delay(THIRD_CLK_MS);

    PORTC &= ~(1 << PIN_RCLK);
    delay(THIRD_CLK_MS);

    PORTB |= 1 << PB5;
}

```

```
void setup() {  
    Serial.begin(9600);  
  
    // Use onboard LED  
    DDRB = 1 << PB5;  
  
    DDRC = 0x1F;  
    // OE = 0: always enable  
    PORTC = 1 << PIN_CLR_LA | 0 << PIN_OE_LA;  
  
    transmit_data();  
}  
  
void loop() {  
}
```